Chapter 1

INTRODUCTION TO STATISTICS

1.1 INTRODUCTION

Statistics is a branch of applied mathematics that deals with every aspect of the data. It involves the proper method for collection, description, analysis and inference of conclusions from the quantitative data. Now-a-days, with the advancement of science and technology, the scope of statistics has widened to almost every domain of human life such as physical or social sciences like medical science, physics, chemistry, economics, politics, psychology, sociology, weather forecasting, health insurance, traffic monitoring, investments, manufacturing etc. It seems that **Dr. A.L. Bowley** has rightly said that, *"Knowledge of statistics is like knowledge of foreign languages or of algebra; it may prove of use at any time under any circumstances."* And also **H.G. Well's** prediction that, *"Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write"* is rightly proved.

Statistics is based on quantitative information which is more accurate. Quantitative statements contain numbers, figures, tables, charts or numerical data. These statements give precise, clear and exact meaning, so that people are easily convinced by these statements. Such statements are known as **Statistics.** But there are some quantitative statements which are not in themselves statistics. The statement that prices in a specific country are increasing reveals incomplete information about the nature of problem or study. But if figures of prices of various years are provided then we better understand the nature of the problem or study. Moreover these figures can also be used to compare with the prices of some other country. We can also estimate the level of price changes in the future and take measures to keep check on it by framing suitable policies etc.

Often we see that while arguing politicians, debaters etc. use statistics for their arguments in order to convince others. Statistics does not mean only figures, numbers, charts, diagrams i.e., numerical data, but also include statistical methods for collection, presentation, analysis, interpretation of statistical data. Thus the *word Statistics refers either to quantitative information or a method deals with that quantitative information.*

1.2 MEANING OF STATISTICS

Statistics has been defined differently by different users. For a common man, statistics means data, facts or measurements. Some people define it as study of figures. Diagrammatic or Graphic Representation of facts is also understood as statistics. Also some consider it as analysis of figures for forecasting or drawing inferences.

However, it is true that statistics is the aggregate of facts expressed numerically, but it is an inadequate description of it as discipline of study.

The word statistics is used in two different senses :

(1) Plural Sense (Statistics as Data) (2) Singular Sense (Statistics as a Science)

Statistics is used for collecting, organizing, presenting, analyzing and interpreting numerical data or data that may be converted into numerical form. It involves variety of methods to assist in the analysis of data for the purpose of decision-making. The important area of the application of these statistical techniques is in business and commerce, which is called study of business statistics.

This book deals with business statistics, so here we discuss how statistical data, tools and methods are used for taking decision in the business to run it efficiently.

1.3 DEFINITIONS OF STATISTICS

There are as many definitions of statistics as there are authors. In 1869 **Quetelet** collected 180 definitions of statistics. *The definitions of statistics can be broadly classified into two categories :*

(I) Definitions in the Plural Sense (as Statistical Data)

(II) Definition of Singular Sense (as Statistical Methods)

In first category, definitions lay emphasis on statistics as data while in second category, it emphasize on statistics as a scientific method.

I. Statistics as Data (in Plural Sense)

Statistics in the plural sense implies as set of numerical facts or figures, systematically collected for some definite purpose under investigation. It refers to collection of numerical data or figures obtained from any field of enquiry such as statistics of prices, food, production, expenditure, income, population, educational institution etc.

Data in numerical form is referred as statistical data, but it should be noted that any set of numerical data cannot be regarded as statistics. It is rightly said that, *"All statistics are numerical data, but all numerical data are not statistics."*

These are certain characteristics which must be satisfied by a given set of numerical data to termed it as statistics. For the purpose of understanding characteristics, we first go through same definitions of statistics in the plural sense :

1. "Statistics are numerical statements of facts in any department of enquiry placed in relation to each other."
   –Dr. A.L. Bowley
2. "Statistics are classified facts representing the conditions of the people in a state ..... especially those facts which can be stated in numbers or in tables of numbers or in any tabular or classified arrangements."
   –Webster
3. "By statistics we mean quantitative data affected to a marked extent by multiplicity of causes."
   –Yule and Kendall
4. "Measurements, enumerations or estimates of natural or social phenomena systematically arranged so as to exhibit their inter-relations."
   –L.R. Conner
5. "Statistics are numerical descriptions of quantitative aspects of things and they take the form of counts or measurements."
   –Wallis and Roberts

All the above definitions cover only some but not all characteristics of statistical data.

By **Prof. Horace Secrist,** *"Statistics may be defined as the aggregate of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated according to a reasonable standard of accuracy, collected in a systematic manner, for a predetermined purpose and placed in relation to each other."*

This definition of **Secrist** can be taken as comprehensive and adequate definition of statistics, since it covers most of the characteristics of statistical data.

1.3.1 Characteristics of Statistics as Numerical Data

Some important characteristics of good statistical data are :

1. **Statistics are aggregates of facts :** Single or unrelated facts do not throw light on any problem. So a single figure related to age, height, weight, birth, death, sale, purchase etc. even though numerically stated cannot be called statistics. But a group of figures related to ages, heights, weights, births, deaths, sales, purchases etc. would be called *Statistics*. For example, a single fact say 500 lakh metric tons production of copper in India during the year 2000 is of no interest to a statistician, as the figure is unrelated and incomparable, but if production figures for several consecutive years are available, then they could be known as *statistics* as the figures are aggregate and comparable.
2. **Statistics are affected to a marked extent by multiplicity of causes :** It means that collection of facts and figures are affected to a considerable extent by a number of forces operating together. It is usually not possible to study the effects of anyone of these forces separately. For example, statistics of production of rice is affected by the rainfall, fertility of soil, quality of seeds and manure, method of cultivation etc. It is very difficult to study separately, the affect of each of these forces on the production of rice. Also, it is true that price level in market is influenced by many factors like demand, supply, import, export, capital etc. and it is difficult to study separately the effects of each of these facts on price level.
3. **Statistics should be numerically expressed :** There are generally two types of statements of facts, quantitative and qualitative. Quantitative statements expressed in numbers but qualitative statements such as 'India is a poor country', 'prices are rising', 'production of wheat is not sufficient', do not express statistics. The reason is that such statements are vague and constitute nothing. So only quantitative statements expressed numerically are statistics.
4. **Statements should be enumerated or estimated according to reasonable standard of accuracy :** Statistical data are collected either by enumeration or estimation. Enumeration of data is actual counting of data. Estimation cannot be as precise and accurate as actual counts or measurements. Enumeration is done only when the field of enquiry and units to be studied are limited e.g., to find the number of students in a college, data is collected by actual country or to find out the weight of students of a particular class, then data is collected by actual measurement. If the scope of enquiry is very wide or the units are very large or the cost of measurements is high, then enumeration is usually avoidable or not possible and in such cases, data can only be estimated by examining only a fraction of the total area of enquiry.

   When statistics are to be obtained by measurement of units, it is necessary to maintain a reasonable degree or standard of accuracy, in measurements. The standard of accuracy depends upon the purpose of enquiry. It cannot be uniform for all types of enquiries. For example, for weighing rice, a few grams add or increase may not make much difference, but for weighing gold, it matters a lot.

According to **Dr. A.L. Bowley,** *"Great numbers are not counted correctly to a unit, they are estimated."* It is normal that the estimated figures cannot be absolutely accurate. Some errors or differences in counting and measurement are likely to take place, but it should be as minimum as possible. The reasonable standard of accuracy will depend upon the nature and purpose of enquiry. **Prof. Secrist** has stated that, *"What are reasonable standards depend upon the purpose which the statistics are to serve."*

5. **Statistics should be collected in a systematic manner :** The statistical data should be collected in a systematic manner to reach at precise and proper conclusion. The data collected in haphazard manner may lead to misleading and fallacious conclusions. According to **Prof. Secrist,** *"Stray and loose bits of quantitative information, hear say and unrelated material gleaned here and there, from indiscriminate sources having no common basis of selection, even when numerical can be termed statistics only by confusion of terms."* A suitable plan of data collection should be prepared beforehand and trained personnel should be used to roll the required data. Also care should be taken to reduce the personal bias and prejudices to minimum, so that reasonable level of accuracy be attained in statistics collection.
6. **Statistics should be collected for pre-determined purpose :** Statistics should be collected for a specific, well defined and pre-determined purpose. A plan of enquiry should be prepared according to such purpose, otherwise it might happen that the unnecessary statistics are collected while the necessary statistics are left out. It would only be wastage of time, money and energy as well as wrong and fallacious conclusions are drawn for undertaken enquiry. For example, if we are to collect statistics of agricultural production, we must decide beforehand, the regions, commodities and periods for which the data is required.
7. **Statistics should be placed in relation to each other :** The statistical data that are collected should be comparable with each other. Comparison is one of the function of statistics. For this purpose, the data must be homogeneous and not heterogeneous. This can be achieved by collecting data in identical manner for different periods or for different places or for different conditions. For example, population of India at a particular point of time can be compared with the population of other countries like China, U.K., U.S.A. etc. (is case of homogeneous data). Heterogeneous data like marks of students of a particular class and height of trees, can never be placed in relation to each other and thus cannot be compared for analysis.

In the absence of the above characteristics, numerical data cannot be called statistics and hence we can say *"all statistics are numerical data but all numerical data are not statistics."*

II. Statistical Methods (in Singular Sense)

Numbers play an essential role in Statistics. They provide data as raw material, which is to be processed through the use of appropriate statistical methods. So, statistics in the singular sense is a science, which provide statistical methods that can be used to organize, present, analyse and interpret the information effectively.

Therefore we refer the term *"Statistics"* as the subject of statistical activity dealing with the theories and methods of collection, analysis and interpretation of data. It is applied for handling numerical data in all branches of business, commerce, trade, science or human activities where measurement or counting is possible. These methods finally help in taking decisions on the face of uncertainty and testing hypothesis.

To know about the various aspects of statistics, some of the *important definitions of the science of statistics are given below :*

1. "Statistics may be called the science of counting." **–A.L.Bowley**
2. "Statistics may rightly be called the science of averages." **–A.L. Bowley**
3. "Statistics is the science of the measurement of social organism, regarded as a whole in all its mainfestations." **–A.L. Bowley**
4. "Statistics is the science of estimates and probabilities." **–Prof. Boddington**
5. "Statistics may be defined as the collection, presentation, analysis and interpretation of numerical data." **–Croxton and Cowden**
6. "Statistics is the science which deals with the methods of collecting, classifying, presenting, comparing and interpreting numerical data collected to throw some light on any sphere of enquiry." **–Seligman**

Most of the above definitions are inadequate definitions, but **Croxton and Cowden** has given a very simple and concise definition of statistics.

This definition clearly points out four stages in a statistical investigation namely :

(i) Collection of data.
(ii) Presentation of data.
(iii) Analysis of data.
(iv) Interpretation of data.

However one more stage may be added to above stages as **'Organization of data.'**

Thus statistics may be defined as the science of collection organization, presentation, analysis and interpretation of numerical data.

1.3.2 Stages in Statistical Investigation

According to above definition there are five stages in statistical investigation.

1. **Collection of Data :** Under this stage, the relevant data is collected from various sources. But utmost care should be taken while collecting data. Data should be taken in a planned manner. Without proper planning, the collected data may not be suitable for the purpose of enquiry and lot of time and money may be wasted. The collected data should be precise, exact and meaningful. The data may be available from primary or secondary sources. Data is said to be primary when it is collected originally by an investigator or agency and it is said to be secondary when it is collected from published or unpublished sources, which have already collected data.
2. **Organisation of data :** Data collected by investigator is either primary or secondary. Generally data collected from published sources (i.e. secondary) is in organised form, but still investigator may have to reorganize it according to his needs. However a huge mass of figures that are collected from a survey i.e. primary data, need compulsory organisation.

The organisation of data is done in three parts :

(a) *Editing* : Editing is done for detecting error, inaccuracies, irregularities, inconsistencies in the data collected

(b) *Classification* : The purpose of classification is to arrange the data into same groups or classes according to some common characteristics possessed by the items constituting the data. Unclassified data is of no use. **J.R. Hicks** has rightly said, *"Classified and arrange facts speak themselves, unarranged – they are as dead as mutton."*

(c) *Tabulation* : The last step in the organization of data is tabulation. The purpose of tabulation is to arrange the data in rows and columns so that there is absolute clarity in the data presented.

3. **Presentation of Data :** After the data have been collected and organized, it is ready for presentation. Data presented in systematically and precise manner make statistical analysis easily understandable. The organised data can be presented with the help of statistical tables, diagrams, graphs etc. It includes tables, bar graphs, pie-charts, flow charts etc.
4. **Analysis of data :** After collection, organisation and presentation, the next steps is to analyse data. The main objective of analysis is to prepare data in such a fashion which helps the investigator to dig out useful information to arrive at certain definite conclusions. The methods most commonly used are *Measures of Central Tendency, Measures of Dispersion, Skewness, Correlation, Regression, Interpolation etc.*
5. **Interpretation of Data :** Interpretation or drawing valid conclusions on the basis of data analysed is the last stage in a statistical investigation. It requires high degree of skill, experience and common sense. If the analysed data is not interpreted carefully, it may lead to wrong and fallacious conclusions, which further defeat the basic objective of enquiry.

1.4 FUNCTIONS OF STATISTICS

Statistics is used to present the numerical data in such a form that is easily understandable by human mind and used easily to make comparisons, derived valid conclusions etc. **Robert W. Buges** describes the functions of statistics in these words, *"The fundamental gospel of statistics is to push back the domain of ignorance, prejudice, rule of thumb, arbitrary or premature decisions, traditions and dogmatism and to increase the domain in which decisions are made and principles are formulated on the basis of analysed quantitative facts."*

The main functions of statistics are as follows :

1. **To Present Facts in Numerical Figures :** One of the most important function of statistics is to present the facts relating to given phenomena in numerical figures. Numerical presentation helps in having a better understanding of the nature of problem. Facts expressed in words are often vague and are likely to be understood differently by different users.
2. **To Present Complex Facts in a Simplified Form :** The raw data is often unwieldy and complex. The function of statistics is to simplify a large mass of complex data and present it in a more understandable and intelligible form. **Dr.**

Bowley states that, *"The object of a statistical estimate of complex groups is to present an outline to enable the mind to comprehend with a single effort the significance of the whole."* In the words of **W.I. King**, *"It is for the purpose of simplifying these unwieldy masses of facts that statistical science is useful. It reduces them to numerical totals or averages which may be abstractly handled like any other mere numbers. It draws pictures and diagrams to illustrate general tendencies and thus in many ways adapt these groups of ideas to the capacity of our intellects."*

The simplification is achieved by the summarization of data so that broad-features of the given problem are brought into our focus. Statistical technique like average, dispersion, graph, diagram etc. make the numerical data more understandable.

3. **To Enlarge Individual Knowledge and Expertise :** Statistical methods provide us such a master key that we can use these methods anywhere and can study any problem in its correct context and prospective. Statistics enlarges human knowledge and expertise in the solution of various problems. In the words of **A.L. Bowley,** *"The proper function of statistics, indeed is to enlarge individual experience."* In the words of **Whipple,** *"Statistics enables one to enlarge his horizon."* Many fields of knowledge would have remained closed to mankind foreever without the efficient and useful techniques and methodology of the science of statistics. The concepts such as national income, population, agricultural production, industrial production, poverty, unemployment etc. can be understood clearly and precisely using statistical techniques.
4. **To Provide Comparative Study :** The science of statistics does not merely mean counting but also comparison. Comparison is easy and in meaningful quantitative terms. Simplified data gives us a helping hand in making comparative study of two or more phenomena. No conclusions can be drawn without comparisons. It is said that everything is understood in a relative sense. **Dr. A.L. Bowley** rightly states, *"A chief practical use of statistics is to show relative importance, the very thing which an individual is likely to misjudge. Statistics are almost always comparative."* Statistical techniques of averages, variances, coefficients, ratios, percentages, trends, graphs and diagrams offer the various means of comparison.
5. **To Study Relationship Between Different Facts :** Statistics helps to investigate whether the two or more phenomena are correlated. It also find out the degree of the relationship and nature of relationship (i.e. positive or negative). For example the relationship between demand and supply, income and expenditure, rainfall and agricultural production, age of husbands and their wives etc. can be studied with the help of correlation anaysis. Also, techniques of regression analysis helps to find mathematical relation fitted in the given data.
6. **To Formulate Policies in Different Fields :** Statistics helps in formulating policies in various fields, specially in social, economic and business fields. Various laws have been developed on the basis of statistics. **e.g. Malthus,** *"Theory of Population"* and **Engel's** *"Law of Family Expenditure"* are due to the statistics collected. Policies in every field, whether private or government, are formulated on the basis of statistical data. For example government policies regarding taxation, prices, investments, imports and exports, social security, etc are guided by

statistics and also a firm can know the tastes and preferences of the consumers and decide to make its product accordingly. **Vital statistics** comprising birth and mortality rates help in assessing future growth in population and this information is very useful in designing any scheme of family planning.

7. **To Help in Forecasting :** The success of planning and policies formulation by the government or a business depends largely on its accuracy in future forecasts. Statistics provides scientific techniques for extrapolation and forecasting. Knowledge of future trends is helpful in framing plans. For example, forecast can be made with regard to changes in future prices, food production, supply of manpower, population growth etc. results in **five year plans** in India. Various techniques used for forecasting are time series analysis, regression analysis, extrapolation theory etc.
8. **To Measure the Effects :** Statistics helps not only in formulation of policy but also in measuring the effects of the policy implementation. The effects of a change in the rate of interest in banks or G.S.T., Sales Tax etc. can be thoroughly studied with the help of statistics.
9. **To Classify Data :** Statistics helps in classification of data into some groups or classes according to some common characteristics between them. It is necessary for analyzing and interpretation of data for drawing conclusions easily with respect to the phenomena under study.
10. **To Formulate and Test Hypothesis :** Statistical methods are extremely helpful in formulating and testing hypothesis and thereby developing new theories. A hypothesis is a statement about same characteristic of population (or universe). For example, hypothesis like the particular dice is fair or not, whether average height of students of a particular college is 5.8' feets or not, etc. It is possible to test the validity of above statements by using statistical techniques.
11. **To Provide Techniques for Making Decisions Under Uncertainty :** Many times, we face a situation of uncertainty or we have to decide one alternative from many ones.

For example, a businessman might face the situation of uncertainty in choosing an investment plan, to start production of new product, to decide the price for his new product etc. The answer to such problems are provided by the statistical techniques of decision making under uncertainty.

1.5 USES AND IMPORTANCE OF STATISTICS

SCOPE

In the words of **H.G. Wells,** *"Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write."* These words are proving true in present day life. Today it is perhaps difficult to imagine a field of knowledge which cannot use statistics. **M.J. Moroney,** writes, *"Whoever you are, if your work calls for interpretation of data, you may be able to do without statistics, but you doesn't do so well."*

It is impossible to understand Psychology, Sociology, Economics, Finance and Physical Sciences even at their basic stage, without understanding the meaning of various statistical techniques like average, dispersion, correlation etc. Knowingly or unknowingly, every person employ statistical methods in taking many decisions. For example, if a person wish to purchase a laptop, then he studies the pricelists of various companies for making a choice. He uses the idea of range (measure of dispersion) within which the prices vary. The

whole business of insurance rests on statistical facts. That is why **Tippet** remarked, *"Statistics affects everybody and touches life at many points."* It is due to statistics, that many fields have progressed in the past and are still progressing. Statistics has embraces all sciences and diversified fields such as agriculture, industry, psychology, sociology, biometry, geology, planning, economics, business, commerce, management, insurance, accountancy, auditing etc.

We shall discuss briefly the importance of statistics in the following major area :

1. Importance of Statistics to the State and Administration :

We know that the subject of statistics originating for helping the ancient rulers to estimate their military and economic strength. But gradually its scope was enlarged to solve other problems relating to political activities of the state. Now in modern era, the states have grown and the concept of welfare state has taken roots to undertake the responsibilities of social and economic welfare of its people. For this a lot of statistical data is required to know about the social and economic conditions of its people. These statistics help in framing suitable policies and its implementations. Statistics helps the government during both peace and war time. It is observed that the government is the biggest collector and user of statistical data in most of the countries. All ministers and departments of the government like Finance, Defence, Transport, Railways, Food, Post and Telegram, Commerce, Agriculture, Small Scale and Large Scale Industries etc. depend heavily on statistical data for their efficient functioning. For example, Railway department cannot solve the problem of rail transport in Northern India unless it knows how many trains are operating at present, how many are required in total and how much trains be added additionally to selected routes. The most important function of the state is to prepare annual general budget which can be formulated only on the basis of statistics. The reports of government committees and commission appointed for specific purpose are also based on statistics. Moreover, in a democratic form of government, various political parties are also guided by the statistical analysis regarding their popularity in the people. Thus it can be stated that statistics is vital for the state.

2. Importance of Statistics to the Business, Industry and Commerce

With the gradual increase in Industrialization and consequent expansion of trade, businessmen can no longer depend on the old system of working and leave their future on chance. Business is regarded as a profession of forecasting which requires high degree of skill to recognize the pulse of future. Success in business depends upon the precision in forecasting. A businessman or industrialist must make a proper and scientific plans, prepare themselves for competitive markets, properly analysis the past record in order to forecast the future business conditions. The traders and manufacturers have to depend on a variety of factors e.g., past sales records, present labour conditions and prices of raw materials and similar products available in the market etc. All these factors must be taken in account before fixing the price of new product.

Every manufacturer must know in advance 'how much is to be produced', 'how much raw material and workers are needed to produce the estimated quantity' and 'what should be the quality, type, size, colour, grade etc. of the product to be manufactured.' etc. He has to employ statistical techniques in estimating trend of prices, trend of economic activities etc. **Buddington** says, *"In order to succeed in any business today, the businessman must study all the factors which enter into production, buying and selling, exporting and importing of goods in which he deals."* He further says, *"A successful businessman is one who*

estimates most closely approach accuracy." Accuracy can be achieved only by proper analysis of past records with the help of statistical methods. Even the manufacturers, producers, industrialists take decision about the quantity of future production on the basis of past records and trends.

With the growing size and ever-increasing cut throat competition, the problems of the business, industry and trade sectors are becoming complex and they are increasing the use of statistics in decision making. Statistical methods are applied in market and product research, investment policies, quality control of manufactured products, economic forecasting etc. Statistics are equally important to the stock and share brokers, investors etc. Statistical methods of analysis are helpful in the marketing function of an enterprise. It helps in market research, advertisement campaigns in comparing sales performance.

Some of the statistical techniques applied in business, commerce and industry are :

(i) To find main characteristics of available records, the various statistical measures of average, dispersion, skewness, correlation etc. are applied. This calculations facilitates the comparison between different time periods or geographical areas.

(ii) **'Time series'** analysis helps in isolating the components of variation noticed in chronological records of prices, sales, profits, production etc. Various trends like seasonal, cyclic helps in future forecasts.

(iii) **Index numbers** of price, production, cost of living, business activities are known as 'business barometers.'

(iv) **Regression analysis** helps in establishing a relationship between various economic factors e.g., demand and per capita income, production and sales etc.

(v) **Statistical quality control** is now being used in industry for establishing quality standards for products and in the maintenance of quality during production.

We may conclude that industry and business involve risks and it is better to calculate risk, so that precautions can be taken before hard. With the help of statistical tools, these risks can be calculated with high degree of accuracy.

3. Importance of Statistics in Economics

The interaction between statistics and economics was first observed by **William Petty** in his book, **'Political Arithmetic'** in end of 17th century. But effective use of statistics in formation of economic theories and policies took a long time. Theories were based on deductive logic only and also the statistical techniques were not as much advanced to be used in other disciplines.

In year 1890, **Prof. Alfred Marshall** observed that, *"Statistics are the straws out of which I, like every other economist, have to make bricks."* This proves the significance of statistics in economics. Economic problem always involve facts that are capable of being expressed numerically e.g., prices of commodities, saving and investment of income, bank deposits, manufacturing output, volume of trade, agricultural wages etc. Statistical data and statistical methods are of immense help in the understanding economic problems and formulating the economic policies. Without statistical data, no economic problem can be studied. Economists collect and analyse statistical data about economic problems and apply statistical techniques to formulate the policies and arrived at certain decisions. Most of the laws and principles in economics like *Laws of Demand and Supply, Consumer Surplus, Law of Diminishing Marginal Utility, Law of Diminishing Marginal Returns, the Theory of Related Preference etc.* can only be expressed and understood with appropriate statistical data.

Economics is concerned with production, distribution, consumption, exchange etc. Statistics of production help us to adjust the supply according to demand. Such statistics are the measures of productivity of a country. With the help of statistics of production, progress of a nation can be measured year after year. Census of production has become an important feature of every nation. In the distribution too, statistics play a vital role. Statistics is a tool in the hands of economists with which they compute national income and wealth and its distribution among people of the country. Statistics of consumption enable us to find out the way in which the people of different classes of society spend their incomes. It also helps to know their standard of living and taxable capacity. In the field of exchange, statistics is very useful in studying markets, laws of prices based on demand and supply, cost of production etc.

A statistical approach to an economic problem not only leads to its correct description but also indicates how it is to be tackled. The increasing importance of statistics in the study of economic problems has resulted in a new branch of study called **Econometrics.** In the words of **Tippet,** *"A day might came when the departments of economics in the universities will go out of the control of economic theoreticians and came under the control of statistical workshops, in the same manner as the department of physics and chemistry have came under the control of experimental laboratories."*

4. Importance of Statistics in Planning

"Economic planning means making of major economic decisions – what and how much is to be produced and to whom it is to be allocated – by the conscious decisions of a determinate authority on the basis of a comprehensive survey of the economy as a whole." It implies a conscious and efficient action to attain pre-determined economic aim. Today, is an age of planning, statistics plays a very vital role in planning. Statistics is indispensable in planning .

In the words of **Tippet,** *"Planning is the order of the day and without statistics, planning is inconceivable."* Planning is indispensable for achieving faster rate of growth through the best use of national resources. Using statistical techniques, it is possible to access the amount of various resources available in the economy and accordingly planning, would be done. Statistical techniques and principles can also guide the Government in adopting suitable policies for betterment of economic problems.

The importance of statistics in planning can be explained as follows :

(i) Statistics is essential to draw plans for economic development to achieve the objective of utilizing the economic resources in the best possible ways.

(ii) To know the position of economic development of our country as compared to other countries, statistics are of great help. The problems like unemployment, over-population, lack of industrialization, low rate of capital formation are characteristics of underdeveloped and underdeveloped countries.

(iii) Use of statistical technique in the evaluation of executed plans is necessary. The difference between the largest and achieved result should be worked out.

(iv) Statistics are used to determine priorities in different sectors for development. This can be done by examining the structure and importance of consumption, production, exchange and distribution sectors of the economy.

Economic planning is done to achieve pre-determined targets. They have to be expressed in numerical facts. Financial planning is a part of economic planning. Financial

resources are moved through taxation, public saving, deficit, financing etc. Planning and budgetary are integrated. All these functions can be preformed efficiently with the help of statistics.

For planning, sometimes said that, "Planning without statistics is a ship without rudder and compass."

5. Importance of Statistics in Banks and Insurance Companies

Banks use statistical techniques to decide the amount for reserves that is to be kept with them in cash to meet the requirements of day to day transactions and demand of their customers for money is satisfied. Otherwise public confidence in the banks is lost. Further, various policies of investment and sanction of loans are also based on the analysis provided by statistics.

The business of insurance is based on the studies of life expectancy in various age groups. Insurance companies have to prepare life tables and calculate premium rates. All this work involves the use of statistical principles and methods.

6. Importance of Statistics in Research

Statistics is indispensable in research work. According to remark of **Wallis and Roberts,** *"Statistics is not a body of substantive knowledge but a body of methods for obtaining knowledge."* Most of the advancement in knowledge has taken place due to research conducted with the help of statistics. So statistics is known as the body of methods for obtaining knowledge. The statistical methods simplify the mass of complex data, present facts, facilitate comparisons and help in testing hypothesis, frame policies and make predictions.

1.6 LIMITATIONS OF STATISTICS

Statistics is an extremely useful science in almost all spheres of human life, but it has its limitations and shortcomings too. **Newsholme** remarks, *"It (statistics) must be regarded as an instrument of research of great value, but having severe limitations, which are not possible to overcome and as such they need our careful attention."*

The following are the limitations of statistics :

1. **Statistics deals with numerical facts only :** This limitation of statistics limits the scope of statistics. Broadly speaking, there are two types of facts : (a) Quantitative and (b) Qualitative facts.

   Quantitative facts are capable of being expressed numerically and thus known as numerical facts. Qualitative facts on the other hand, represent the qualitative characteristics like intelligency, honesty, goodwill, colour of eyes, beauty etc. Quantitative data can be analysed and interpreted with the help of statistical methods but not qualitative data. Sometimes, however it is studied by indirect study of such characteristic through their conversation into numerical figures. According to **Prof. Horace Secrist,** *"Some phenomena cannot be quantitatively measured, honesty of resource fullness, integrity, goodwill, all important in industry as well in life generally, are not susceptible of direct statistical measurement.*

2. **Statistics deals with averages :** The laws of statistics are not universally true like the laws of natural and physical sciences. They are true only on an average. The reason for it is statistics deals with such phenomena, which are affected by a

multiplicity of causes and it is not possible to segregate the effect of one factor only. For example, the statement that average wages of workers in a certain factory is ₹2000 per month, does not necessarily mean that every worker is paid the same amount.

3. **Statistics does not study individuals :** Statistics deals with the aggregate of facts rather than the individual items. It deals with only mass data and throw light on the characteristics of the entire group. For example, the wage earned by a single worker of a factory is not statistics, but the wages of workers of a factory can be used statistically. In the words of **W.I. King,** *"Statistics from their very nature of subject cannot and will never be able to take into account individual cases. When these are important, other means must be used for their study."*
4. **Statistical results are only approximately true :** Most of the statistical studies are based on sampling method of investigation. According to **Boddington,** *"Statistics are estimates and probabilities."* The data collected may also be based on estimates and probabilities. Whether statistical results are based on sample or census data, they are bound to be true only approximately. Thus statistics fails in the fields where perfect accuracy is desired.
5. **Statistics give wrong information without references :** Statistics must take care of references while analyzing and interpretating the results. If references are not considered while interpretating statistical data, the interpretation may be taken wrongly. For example, if salary of X, Y and Z is ₹1,00,000 each, then salary figure shows that each of them is equally paid. But X's income is monthly, Y's income is half yearly and Z's income is annually. Then the above interpretation gives erratic conclusion.
6. **Statistics is only a means :** Some scholars believe that statistics is only a means and not an end in the solution of any problem. The accuracy of conclusions depends upon the intention of the investigator. If he is partial or have any selfish interest, then it affects the conclusion.
7. **Statistics should be used only by experts :** Statistical methods are a double edged weapon and must be handled by experts only. If any makes a decision with a lack of expertise in it, it may give misleading and dangerous results. **Yule and Kendell** have rightly pointed out, *"Statistical methods are most dangerous tools in the hands of in experts."*
8. **Statistics are likely to be misused :** Statistics may be misused by unskilled, inexperienced or dishonest persons. Statistics involves figures which do not bear on their face the label of their quality. The figures can easily be manipulated or molded for desired selfish motive. It is usually said, *"Statistics are like clay of which you can make a god or devil, as you please."*
9. **Statistics is one of the methods :** According to **Croxten and Cowden**, *"It must not be assumed that the statistics is the only method to use in research, neither should this method be considered the best attack for the problem."* Statistical method is not the only method to study under all circumstances. Many times, this method may not suggest the best solution for the problem. So care must be taken in choosing tools, techniques and test to be applied on the data and should verify the conclusion with the help of qualitative method.

10. **Need of homogenous data :** Statistical methods are useful only when data is uniformed and homogenous, i.e., it should be subject to a stable casual system. There should not be much change in the group of factors responsible for variation in the numerical data. Comparison of data for drawing conclusion cannot be made when quantities are measured in different units. For example, we cannot compare height of tree with price of shirt.

IMPORTANT THEORETICAL QUESTIONS

SHORT ASNSWER TYPE QUESTIONS

1. Define Statistics in 'Plural sense.'
2. Define statistics in 'Singular sense.'
3. Write a short note on the statistics.
4. Discuss the nature of statistics.
5. Give the limitations of statistics.
6. What are the important functions of statistics.

LONG ANSWER QUESTIONS

1. What do you understand by Statistical Methods?
2. Define statistics? Discuss its characteristics.
3. Define statistics with its uses, limitations and functions.
4. Explain clearly the functions and limitations of statistics in the fields of business and economics.

○○○

Chapter 2

COLLECTION AND CLASSIFICATION OF DATA

2.1 INTRODUCTION

Statistical data which means facts or information are the basic raw material of statistics. For any statistical inquiry, whether related to business, economic or social sciences or related to government/private sector, the basic problem is to collect facts and figures related to that particular field of inquiry.

The person who conducts (i.e., counts or measures data or figures individually) the statistical inquiry for further statistical analysis is known as ***investigator***. Generally, an investigator must be trained and efficient statistician. The persons from whom the information is collected are known as ***respondents*** and the objects on which the measurements are taken are called ***statistical units***.

Statistical investigation is a comprehensive process and it has to pass through different stages from collection of data to the submission of final report, like (a) *Planning the statistical investigation,* (b) *Collection of data* (c) *Editing and presentation of data* (d) *Analysis and Interpretation of data* and lastly (e) *Preparation of final report.*

Before we discuss about the *collection of data* for statistical inquiry, firstly it is necessary to examine prerequisites for it.

2.2 PREREQUISITES FOR COLLECTION OF DATA

A statistical investigation is a process of collection and analysis of data. The quality and accuracy of investigation depends on the preparation made before the inquiry is conducted. A properly planned investigation can give the best results in least time and cost.

The following points should be considered before commencing the collection of data :

1. Objective, scope and nature of the inquiry.
2. Statistical units to be used.
3. Source of information.
4. Methods of data collection.
5. Degree of accuracy desired.

1. Objective, Scope and Nature of the Inquiry

The first and foremost essential step in organizing any statistical inquiry is to define the purpose of the investigation in precise and clear form. According to **Johnson and Jackson,** *"The real purpose of statistical methods is to make sense out of facts and figures to prove the unknown and to cast light upon the situation."*

A statistical inquiry may be either for general purpose or for a special purpose. In ***general purpose*** inquiry, the obtained data is useful for several purposes e.g., data of population census in India help in determining the number of males or females, literates or

illiterates; employed or unemployed, distribution of income, distribution of age etc. A ***special purpose*** inquiry contains data which is useful in analyzing a particular problem only like height of students of B.Com. II class in a particular college etc.

In the words of **Prof. W.A. Neiswanger,** *"Statement of objective is of basic importance because it determines the data which are to be collected, the characteristics of the data which are relevant, relations which are to be explored, the choice of the technique to be used in these explorations and the content and the form of the final report."*

The objective of inquiry is to facilitate the classification, tabulation analysis and interpretation of the collected data. Even a slight variation in the purpose of inquiry may require entirely or partially different statistical data. Moreover, failure to set out clearly the purpose of inquiry results in confusion and waste of resources. Opinion of **Prof Yo-Lun-Chou** about the objective of inquiry states, *"If he fails to do this, the data collected may be entirely irrelevant or may even tend to cloud rather than clarify the problem. It is well to remember that the quality of statistical conclusions depends upon the appropriateness and accuracy of the data, which in then depend upon exactness in formulating the problems. Statistical techniques, no matter how refined and precise, cannot yield useful results for arriving at decisions on problems, if they are not applied to appropriate data."*

After specifying objective, it becomes easy to determine the nature of statistics (data) to be collected and statistical methods to be employed for the analysis of the data.

The scope of inquiry specifies the area of investigation and the type of data to be collected for the achievement of given objective. For example, to study the standard of living of people, or problems related to industrial disputes, wages of workers etc., the inquiry may relate to India as a whole, or one particular state or an industrial town.

Three factors that influence most on scope of inquiry are objective of inquiry, availability of time and resources.

Time to complete the process of investigation should neither be too large or too small. Larger the period of investigation, the more variations in data is expected as the data may be out dated or rendered useless for the purpose of the inquiry in hand. Also, in the absence of precise specification of the scope, it is possible that irrelevant data may be collected and relevant data are left out. This results in wastage of time, money and other resources.

In the words of **Prof. A.R. Ilersic,** *"In practice, however the difficulties of actually collecting data are often not as great as the difficulties which arise at the outlet in defining the scope of the inquiry. There is first the problem of deciding exactly what information the statistician wants, then the best way of getting it."*

2. Statistical Units to be Used

Statistical data are obtained by observations of different items or the *units of investigation.* The unit in which the investigator counts or measures the data selected for enumeration, analysis and interpretation is known as ***'Statistical Unit'***. A unit of investigation, may be a person or a house, or a family, or an institution or a physical units of measurement such as metres, kilometres, kilograms, quintals, hours, days, weeks etc. which are well defined and form the basis for the collection of data. In the words of **Wheldon,** *"A statistical unit is an attribute or a set of attributes conventionally chosen so that individuals or objects possessing them may be counted or measured for the purpose of statistical inquiry."*

The mistake in the selection of the unit is more harmful than mistakes in the collection of data. Seeking this, the following characteristics must be possessed by statistical units :

(i) It must be suitable for the purpose of enquiry.

(ii) It must be simple and clear to understand and unambiguous.

(iii) It must be definite, specific and appropriate for the enquiry.

(iv) It must be stable in character, because if unit changes its character like firstly use yards, than using metres, then the data will lose the comparable value.

(v) It must ensure homogeneity and uniformity throughout the investigation so that there can be valid comparisons. Also if units are defined differently at different stages of the inquiry, the comparisons become difficult and wrong conclusions will be drawn.

3. Sources of Information

Having decided about objectives, scope and statistical units to be used, now next we have to decide about the sources from where the information (data) can be collected or obtained. The sources of the data may be :

(a) Internal and External : The sources of information can be internal or external. The use of either or both will depend on the nature and scope of inquiry as well as availability of internal information. Internal data of any organization (business, firm, company etc.) are collected by the organisation itself for internal purposes like data of production, sales, profits, imports and exports, loans, capital employed etc. The use of internal sources usually need not formal inquiry, but sometimes the task of inquiry is given to consultancy firm or chartered or cost accountants who give their views (conclusions) on the basis of internal data. Generally, a statistical inquiry is conducted when sources of information are external to the organization. External data are obtained from the publications of some other agencies like Government (State or Central), International Organisations, Private Research Institutions etc.

(b) Primary and Secondary : When the data are collected originally i.e. by observing the units of the problem under investigation, it is called ***primary data***. On the other hand, data obtained from some published sources such as publications of Government, Semi-Government organizations, magazines, periodical, research journals, newspaper, budget etc. or already collected and used by some other agencies is called ***secondary data***.

Both the sources have their own merits and demerits. While collecting primary data, the type of inquiry, objective, scope, statistical units of the data are kept in mind, but no such things apply on secondary data, as the data have been collected already. In secondary data, care should be taken in editing, scrutinising of data with respect to its suitability and reliability. On other hand, the data which are primary in the hands of one person, may be secondary in the hands of other person. The selection of particular source is dependent upon many factors like :

(a) Purpose of Inquiry.

(b) Time required

(c) Accuracy desired.

(d) Availability of funds (money).

(e) Availability of other facilities.

(f) Nature of the investigator.

4. Methods of Data Collection

There are various factors that determine the methods of data collection. The accuracy and the cost of collected data varies with the method of inquiry. It depends upon these factors that whether sample inquiry is to be conducted or a complete count or measurement is to be taken. Broadly there are two methods of data collection :

(a) Census : Census is a complete enumeration of each and every unit under investigation.

(b) Sample : The information collected by observing only a part of the complete units is called sample and conclusion drawn from sample basis is concluded as the conclusion of whole data.

Census method is costly, time consuming and using more manpower as compared to sample method. The choice between both the methods depend upon :

(i) availability of resources, (ii) availability of time, (iii) degree of accuracy desired, (iv) nature and scope of the inquiry etc.

Both the methods have same merits and demerits.

Merits of Census Method

1. Data is obtained from each and every unit of population.
2. Results obtained are mostly representative, accurate and reliable.
3. It is an appropriate method of obtaining information, that is why the population data is collected by conducting a census in generally every ten years by census method in all over the world.
4. Data collected by census method can be widely used as a basis for many other purposes too.

Demerits of Census Method

1. Complete enumeration requires a lot of time, money, manpower etc.
2. It is difficult to collect the data by census method practically. Only big business, organisation and governments have sufficiently large resources to afford their method.
3. In case of urgent requirement of data, this method is useless.
4. It loses its worthiness, if a sample can yield equally reliable results.

Merits of Sample Method

1. It reduces the cost and time of inquiry.
2. In the absence of complete availability of whole data, this method remains the only way.
3. If population (universe) is too large, then census technique is not possible, so only way is to apply sample method.
4. Census method requires a huge team to administered for inquiry, but sample method can work with less number of investigators.
5. With a small data to investigate, there are fewer chance of statistical errors.

(6) Articles published by reputed authorities on different subjects.

Primary Data

In a statistical investigation when a suitable, reliable and adequate data is not available then the investigator has to collect the data himself for the purpose of the enquiry. The data which are originally collected by an investigator or an agency for the first time for any statistical investigation and are used in the statistical analysis are termed as primary data. Hence *primary data* are measurements recorded as part of an original study. The collection of primary data is costly and laborious. But the chances that there are inadequacies, shortcomings or pitfalls at the very source of the data are less.

It may be noted that the data which is primary to an investigator may become secondary to someone else. The term primary or secondary is a matter of relativity. The same data which is primary on the part of one may become secondary on the part of someone else.

Choice between Primary and Secondary Data

A proper choice between primary or secondary data needed for any particular statistical investigation is to be decided during planning, after taking into account the nature, objective and scope of the enquiry, the availability of time and money ; the degree of accuracy desired etc. As stated above, primary data are usually collected if there do not exist any secondary data suited to the investigation under study. In some of the investigations both primary as well as secondary data may be used. There is lot of difference in the methods of collection of primary and secondary data. In collection of primary data, the entire scheme of the plan starting with the

definitions of various terms used, units to be employed, type of enquiry to be conducted, extent of accuracy aimed at etc., is to be formulated whereas the collection of secondary data is in the form of mere compilation of the existing data.

Methods of Collecting Primary Data

The methods commonly used for the collection of primary data are :

(1) Direct personal investigation.

(2) Indirect oral investigation.

(3) Information received through correspondents.

(4) Mailed questionnaire method.

(5) Schedules sent through enumerators.

(1) **Direct personal investigation.** This method consists in collection of data personally by the investigator (organizing agency) from the sources concerned. In other words the investigator has to go to the field personally, contact the respondents, convince them in giving the information under study. This way he should seek the information from the respondents. This is a suitable method where the area of coverage is small and the number of respondents to be contacted is small.

Merits : (*i*) Most people are willing to supply information on personal approach. So the response is more encouraging.

(*ii*) The first hand information obtained by the investigator himself is bound to be more reliable and accurate since the investigator can extract the correct information by removing the doubts, if any, in the minds of the respondents regarding questions. In case, of doubt about any information by a respondent, the investigator can tactfully cross questions and examine the correctness of the information.

(*iii*) The data obtained from such investigation is generally reliable and the desired level of accuracy can be attained, because the very person who decides these factors is present.

(*iv*) In a delicate situation of asking some sensitive questions, the investigator can use his skill, tact and experience in avoiding the sharp reaction of a respondent.

Demerits : (*i*) This method is not suitable when the area of coverage is considerably wide or many respondents are to be contacted. In other words this method is not appropriate in an extensive study.

(*ii*) Since a single man has to manage to contact all the respondents, the time required is more. Sometimes the enquiry may not get completed within the specified time because of the delay in collecting the data.

(*iii*) The entire success of this method depends upon the investigator. As mentioned above, the investigator should be experienced, tactful,

intelligent, diplomatic and so on. But the absence of one or more of such qualities will lead to collection of unsatisfactory and unreliable data.

(*iv*) If the investigator is biased or prejudiced then the enquiry is simply driven at a predetermined decision.

(*v*) If the investigator acts according to his whims and fancies the findings of the enquiry may become unreliable.

(2) **Indirect oral investigation.** Imagine that an investigation is being conducted to study the habit of drugs among the students of an educational institution. In this case the respondent may not give the true account of the situation. In such a case the 'direct personal investigation' is not practicable because of the unwillingness or reluctance of the respondents to provide the required information. To get the information, the investigator has to contact the persons who are close to the respondent like friends, neighbours, relatives, or the persons who know the information. The investigator takes these persons into confidence and interviews them. Through various questions he tried to find out the facts about the respondent. This method of collecting the data is known as indirect interview method. This type of enquiry is usually conducted by the police department and enquiry committees appointed by the government.

Merits : (*i*) When the direct personal enquiry method fails, this alternative method becomes convenient.

(*ii*) This method is free from the bias and prejudice of the respondent.

Demerits : (*i*) The achievement of the success of this method depends on the tact and efficiency of the investigator. If he can properly handle the persons supplying the information, only then, correct data can be collected.

(*ii*) In contacting few persons in collecting the information of a single respondent, considerable time may be lost. This may also prove to be a costly method.

(*iii*) The information collected may not be fully true. The people who give the information, in their own interest, may colour the information. The choice of a wrong person may contradict other information and this may create confusion. Hence, the accuracy of the data collected, depends to a large extent on the nature and quality of the people from whom the information is obtained.

(*iv*) If the information is based on the data supplied by a single person the true information may not be collected. Hence contacting many persons becomes necessary.

(3) **Information received through local correspondents.** When the information is needed from different places and for a long period of time the

investigator appoints the local correspondents at different parts of the field of enquiry. These correspondants are trained to collect the information from the respondents. They regularly collect the data in their jurisdiction, and send them to the investigator. For example, a textile industry may be interested in the demand forecast of variety of cloths. The correspondents are asked to collect the information of different types of cloth and quantity purchased by some selected families or quantity sold in some reputed shops, every month. These data are sent to the investigator. This technique of data collection is usually employed by different departments of the government, when the data is required regularly. Majority of newspaper and periodicals need the information concerning politics, business, prices, agricultural and industrial output, stock market etc. regularly and hence they follow this method.

Merits : (*i*) The data can be collected economically. If the correspondents are taken as part time workers, to that extent the cost reduces.

(*ii*) Form a wide area data can be collected in a short period.

Demerits : (*i*) If the correspondents are not properly trained, the data collected may not be reliable.

(*ii*) It is observed in some cases that, after some period of time the correspondents do not contact the respondents, but whatever information they feel proper, is sent to the investigator. This will defeat the very purpose of the data collection.

(*iii*) The personal prejudices and whims of the correspondents result in collecting unreliable data.

(4) **Mailed Questionnaire Method.** A questionnaire is a set questions pertaining to the enquiry. It is the list of the questions the investigator is supposed to ask the respondents. From the point of clarity the operational definitions of various terms and concepts used in the questionnaire are also added. Space is provided, to fill the questionnaire, so that the respondents can write their answers. Many of the questions may be objective type, that is some suggested answers are provided to these questions. A respondent can select one or more of these as an answer to the question.

This method consists in preparing a questionnaire which is posted to the respondents with a request for a quick reply. A very polite covering note, explaining in short the aims and objectives of collecting the information, is also attached. Respondents are requested to extend their full cooperation by returning the questionnaire duly filled in time. Respondents are assured of the secrecy of the information supplied by them. Usually a

self addressed, stamped envelope is enclosed for returning the questionnaire and post it back to the investigator. In this way the information is gathered from the respondents.

Merits : (*i*) Respondents from a wide geographical area can be covered without increasing the cost, time and manpower. Hence is considered as an economical method.

(*ii*) This method is used for extensive enquiries covering very wide area.

(*iii*) Every question is interpreted by the respondent in his own way. Hence the personal biases of the investigator or enumerators are not present.

Demerits: (*i*) This method is suitable only where the respondents are educated.

(*ii*) Even if the respondents are literate, they may not be responsible enough to fill the reliable information and send back the questionnaire in time. It is observed that in many of the investigations the respondents, even though are educated, are indifferent in such enquires, unless there is direct or indirect benefit they derive or they are compelled to answer under law. It is possible that a large proportion of the responded may supply puzzling, incomplete, inconsistent and nuclear information which does not serve much purpose.

(*iii*) The questions may be misinterpreted and there is no one to give the idea about what exactly a question aims at. This may lead to misinterpretation of the questions and collection of unwanted data.

(*iv*) People may not like to put in black and white the information regarding income, amount of sales, number of workers, the working conditions, their likings or disliking etc. It becomes very difficult to verify the correctness of such suppressed information.

(*v*) There is no scope for asking supplementary questions for cross checking of the information supplied by them.

(*vi*) The follow up work may become necessary in case of incomplete or inconsistent data.

Drafting or Framing the Questionnaire. Drafting a questionnaire is a specific art. A questionnaire should be drafted with utmost care and caution so that all the necessary information for the inquiry may be collected without any difficulty. The success of the questionnaire method of collecting information depends largely on the proper designing of the questionnaire. The general points which may be considered in drafting a questionnaire are :

1. The number of questions in a questionnaire should be minimum. In other words, the questionnaire should be concise and should contain only those questions which would furnish all the necessary information relevant for the purpose. An over enthusiastic investigator may think that more the information collected, through more questions, may be useful at the time of analysis. But the additional information may not be really found useful at the time of analysis. A large number of questions may scare the respondents. Such mistakes are likely to be committed by inexperienced investigators.

2. The questions should be clear, brief, and unambiguous. The interpretation of any question should not become a problem to the respondent. The required information should be clearly aimed at through a very simple language.

3. The questions should be arranged in a logical order. When there are few related questions they should be arranged in a way as to initiate a chain of systematic answers. The sequence of the questions should be such that there should be a logical flow of thought. The logical arrangement of questions will facilitate work of classification.

4. The words which mean differently to different people should not be used. For example, the word capital, prosperity, comfort etc., may mean different to different persons. Such words should be avoided or proper clarification should provided.

5. Questions involving calculations as percentages, ratios etc., should be avoided as far as possible.

6. Questions which force the respondent to recollect from his memory should not asked. People do not like those questions which may tax their memory.

7. Sensitive questions which may irritate the respondents or may hurt their feelings should not be asked. Questions like 'Why you have faith in this religion ?', 'Why your children do not look after you in your old age ?' etc., which might hurt the sentiments, pride or prestige of an individual should not be asked. It is also wise to avoid questions which the respondent may be unwilling to answer.

8. The questions should be as far as possible objective type. Objective type questions are those where some suggestive answers are provided. For example, for the question, "How do you normally reach your office ?", the possible answers may be (*i*) By walk (*ii*) By bus (*iii*) On bicycle (*iv*) On two wheeler (*v*) By car. The respondent will select the one (sometimes more than one), which he feels the most appropriate. The advantage of such questions is that the respondent need not think about what exactly is to be

written. Classification of these answers will become easy. It is observed in many of the surveys that, the objective type questions are usually answered and other questions are left unanswered.

(5) **Schedules sent through enumerators.** This is the most practical method of collecting primary data. Majority of the primary data are collected by this method. The investigator appoints enumerators to collect the data on his behalf. These enumerators are trained about the way they should approach the respondents, the way they should ask the questions, the way the questions should be interpreted, the way the answers should be filled in the schedule etc. (A questionnaire is a list of questions for which the answers are entered by the respondent. While a schedule is the device of obtaining answers to the questions in a form which is filled by the enumerators, as answered by respondents). The enumerators go to the respondents with the schedule, ask them the questions therein and record their replies. The data collected this way is brought back to the investigator.

It is necessary that the enumerators should not blindly go on asking the questions in the schedule. It is left to one's guess the consequences faced by an enumerator who asked, "Are you married ?", the answer was "No" and the next questions the enumerator asked was "How many children do you have ?". To correct the mistake he changed the order of the questions and asked another respondent "How many children do you have ?", the answer was "Two". The question that followed was, "Are you married ?".

Merits : (*i*) The enumerators can convince the respondents is giving the information needed in the enquiry.

(*ii*) The presence of the enumerator may induce the respondents in giving the information. hence the chances of non-response is less, which does not happen in "mailing questionnaire method".

(*iii*) Misinterpretation of the questions can be avoided.

(*iv*) The exactly required information may be gathered avoiding unnecessary information. Such as, in a enquiry, only annual gross income may be enough, but some respondent may give the data about basic pay, dearness allowance, other allowances etc., which are not required.

(*v*) Unlike the 'mailed questionnaire method', this method can be used with advantage even if the respondents are illiterate.

Demerits : (*i*) Compared to other methods of collecting primary data, this method is fairly expensive. The expenditure on training the enumerators, the remuneration paid to them, their expenditure on the conveyance etc., are to be born by the investigator or the agency collecting the data.

(*ii*) Compared to the 'mailed questionnaire method' it is more time consuming method.

(*iii*) Inefficiency on the part of the enumerators coupled with personal bias may adversely affect the results of the enquiry.

(*iv*) If the enumerators are not properly trained or the enumerators are not honest, skillful, diplomatic then the data collected may not be reliable. In some enquiries it happens that the enumerators, do not contact the respondents at all and fill the schedules of their own. Such practice may defeat the very purpose of collecting data.

(*v*) If the drafting of the schedule is not properly done the enumerators may find it very difficult to get the required data or the correct data may not be gathered.

(*iv*) The follow up work may become necessary in case of incomplete or inconsistent data.

Sources of Secondary Data

The Secondary data for an enquiry may be searched in

(1) Official publications of Central Government, State Governments, reports of committees and commissions appointed by the government, etc. We may get the statistical material like crop statistics, trade statistics, population statistics, etc. in the publications of information by the government.

Discrete Frequency Distribution

In a statistical data when the values are largely repeated then we can prepare a table showing the different values of the variable along with the number of repetitions of each value. Such a table is called a discrete frequency distribution or discrete frequency table. To prepare this type of distribution, we have just to count the number of times a particular value is repeated which is called the frequency of that value. Suppose we have 100 values in a ungrouped data. To count the number of times, say 10, is repeated, we have to go through all the 100 values and count how many times 10 is present. Then, to count another value, say 15, once again we have to go through all the 100 values. For every value of the variable, we need to scan all the values of the data. To avoid this and to facilitate counting, place all possible values of the variable, preferably, from the lowest to the highest in a column. Prepare another column, of 'tally marks'. A tally mark or a tally bar or a tally score is a small vertical bar representing a value of the variable. If the first value, in the ungrouped data is, say 25, put a vertical bar in the second column against the number 25 of the first column. If the next value in the ungrouped data is 50 then put a tally mark against the number 50 in the frequency distribution. To facilitate counting, when there are five consecutive tally marks, the fifth tally mark is put across the earlier four tally marks to make a block of five tally marks. Some space is left in between two blocks. When all the values of the ungrouped data are presented in the table, count the number of bars corresponding to each value of the variable and place it in the column of frequency. This procedure is explained through the following small example.

Continuous Frequency Distribution

In the data pertaining to a continuous variable, normally the values do not repeat. Suppose, we have the data of percentage of marks of 100 students in a class. If the percentages are calculated for two decimal places, we may have all the 100 values different. In such a case a discrete frequency distribution is not appropriate. These figures will have to be grouped into different classes. The distribution of marks may look as follows :

Marks	:	0–20	20–40	40–60	60–80	80–100
No. of students	:	8	25	44	15	8

The above frequency distribution is called a continuous frequency distribution. There are five classes. 20–40 is called the second class. In this class, 20 is called the lower class limit and 40 the upper class limit. The values 0, 20, 40, 60 and 80 are lower limits and 20, 40, 60, 80 and 100 are the upper limits of the respective classes.

The following technical terms are important when data are classified according to classes.

(*i*) **Class frequency.** The number of the observations corresponding to a particular class is known as the frequency of the class. In the example mentioned above, the frequency of the class 40–60 in 44, which implies that there are 44 students whose percentage of marks is 40 or more but less than 60. The total frequency of the five classes is 100 which means that in all there are 100 students.

(*ii*) **Magnitude of the class.** The difference between the upper boundary and the lower boundary of a class is called the magnitude or size or length of the class. The magnitude of the class 20–40 is 20 and the magnitude of the class 15–25 is 10.

(*iii*) **Class mid-point.** The centre of the limits of a class is called the mid-point or mid-value or class marks of a class. It is the value lying half-way between the lower and upper class-limits of a class interval. Mid-point of a class is defined as :

$$\frac{1}{2} \text{ (Lower limit of the class + Upper limit of the class)}$$

For example, the mid-point of the class 40–50 is $\frac{1}{2}(40 + 50) = 45$ and mid-point of the class 35–40 is $\frac{1}{2}(35 + 40) = 37.5$.

Remark. Mid-point of a class can be calculated, more conveniently, as the lower limit of the class plus half of the magnitude of the class. The mid-point of the class 40-50 is $40 + \frac{1}{2}(50 - 40) = 40 + 5 = 45$.

In the construction of a frequency distribution, when the classes are formed, the following points should be taken into account :

(*i*) The classes should be clearly defined. These classes should be non-overlapping so that any value of the data is included in one and only one class. The classes should be taken to cover every value of the data.

(*ii*) The number of classes should neither be too small nor should be too large. With very few classes the frequencies will be crowded in a class leading to a rough classification and loss of information. On the other hand, too many classes will have smaller frequencies of each class. This results in irregular pattern of frequencies. Normally, there should not be less than 5 classes and more than 15 classes in a frequency distribution. However, this depends upon the nature of the data, the desired accuracy during analysis, the spread of the values in the data and the number of values in the data.

(*iii*) The magnitude of the class equals to the ratio of the range of the data (difference between the largest and the smallest value of the data) to the number of classes. As far as possible, the magnitude of the class should be 5 or multiple of 5. From the point of convenience, the class-intervals should be of uniform size. If intervals are not of uniform width, it becomes

difficult to make meaningful comparison between classes. But in the data such as age groups or income groups the classes are taken with unequal magnitudes.

(*iv*) The lower limit of the first class, should be either zero or 5 or multiple of 5.

For example, if the lowest value of the data is 23 and the class-interval is 5 then first class should be 20–25.

(*v*) There are two methods of classifying the data according to class intervals, namely (*a*) "inclusive" method, and (*b*) "exclusive" method.

(*a*) **Inclusive Method.** The classes of the type 30–34, 35–39, 40–44 etc., wherein the upper limit of a class is not equal to the lower limit of the succeeding class, are known as inclusive classes. Under the 'inclusive' method of classification, the values of the data equal to or more than the lower limit of the class and less than or equal to the upper limit of the class are included in the respective class. For instance, the class 35–39 includes all the values from 35 to 39, both inclusive. The next value 40 goes to the class 40–44. The problem with such classes is that, if we come across a value such as 39.2, it cannot be accommodated in any class. Hence such type of classes are suitable when the data the only integer values or the variable is discrete in nature. An example of a continuous frequency distribution with inclusive classes is given below :

(*b*) **Exclusive Method.** The classes of the type 10–20, 20–30, 30–40, etc., are known as exclusive classes. In these classes the upper limit of a class is same as the lower limit of the next class.In the class 30–40 all the values equal to or more than 30 but less than 40 are included. The values which equal to the upper limit of a class are excluded from the respective class.For example, suppose there is a value 30 in the raw data. This value should not be taken in the class 20–30 but, should be included in the class 30–40. The following table is an example of frequency distribution with exclusive classes.

Cumulative frequency distribution. From a given frequency distribution we can derive two frequency distributions called cumulative frequency distributions or derived frequency distributions. Consider the following frequency distribution :

Marks :	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
Students :	5	12	23	29	24	19	10	2

From the above table we can read that there are 5 students who have scored 10 or more but less than 20 marks. Similarly, there are 12 students whose score is 20 or more but less than 30 and so on. From this information we can assert that there are 5 students who have scored less than 20 marks.

The number of students who have scored less than 30 marks is 5 + 12 = 17.

The number of students who have less than 40 marks is 5 + 12 + 23 = 40.

This information can be presented in the tabular form as

Table 5.17. Less than cumulative frequency distribution of marks

Marks	*No. of students*
Less than 20	5
Less than 30	5 + 12 = 17
Less than 40	17 + 23 = 40
Less than 50	40 + 29 = 69
Less than 60	69 + 24 = 93
Less than 70	93 + 19 = 112
Less than 80	112 + 10 = 122
Less than 90	122 + 2 = 124

In the above table the values 5, 17, 40 etc., are derived from the frequencies 5, 12, 23 etc. Hence the name "derived frequency distribution." Also, these are the cumulative sums of the frequencies 5, 12, 23 and so forth. Hence the table can be named as "cumulative frequency distribution." In particular we have calculated the cumulative frequencies for less than the upper class boundaries. Hence the table is known as a "less than cumulative frequency distribution."

Similarly, we can form a table showing the number of students who have scored more than the lower class boundaries. In this case the cumulative sums should be obtained from the highest class to the lowest class. In the above illustration, the number students scoring more than 80 marks is 2, and more than 70 marks is 2 + 10 = 12 etc. These values are tabulated in the following "more than cumulative frequency distribution."

Table 5.18. More than cumulative frequency distribution of marks

Marks	*No. of students*
More than 10	119 + 5 = 124
More than 20	107 + 12 = 119
More than 30	84 + 23 = 107
More than 40	55 + 29 = 84
More than 50	31 + 24 = 55
More than 60	12 + 19 = 31
More than 70	2 + 10 = 12
More than 80	2

The Tables 5.10 and 5.11 together can be expressed as follows :

Table 5.19. Less than and more than cumulative frequency distributions of marks

Marks	*No. of students*	*Less than Cum. Frequency*	*More than Cum. Frequency*
10–20	5	5	124
20–30	12	17	119
30–40	23	40	107
40–50	29	69	84
50–60	24	93	55
60–70	19	112	31
70–80	10	122	12
80–90	2	124	2
Total	124		

Remarks. 1. The less than cumulative frequencies are associated with the upper boundaries.

2. The more than cumulative frequencies are associated with lower class boundaries.

3. The less than cumulative frequency corresponding to the highest class and the more than cumulative frequency corresponding to the lowest class are equal and each equals to the total frequency.

Example 5.13. *Convert the following more than cumulative frequency distribution into a less than cumulative frequency distribution.*

3.1 TABULATION OF DATA

Tabulation is a logical and systemic presentation of statistical data in rows and columns. Rows are horizontal arrangements whereas columns are vertical arrangements of the data. The purpose of a table is to simplify the presentations and facilitate comparisons. It gives the data an orderly arrangement and highlights its basic characteristics. The main purpose of a table is to furnish the maximum information in the minimum possible space, without losing the usefulness and quality of data. Tabulation of data occurs after the classification of data. In fact, tabulation is the final stage in collection and compilation of the data and forms a path to further statistical analysis and interpretation.

3.1.1 Definitions of Tabulation

Some of the important definitions on tabulation are :

"Statistical table is the logical listing of related quantitative data in vertical columns and horizontal rows of numbers with sufficient explanatory and qualifying words, phrases and statements in the form of titles, headings and explanatory notes to make clear the full meaning, context and the origin of the data." **— Tuttle**

"Tables are a means of recording in permanent form the analysis that is made through classification and by placing in juxta position things that are similar and should be compared." **— Secrist**

"Tabulation is the intermediate process between the accumulation of data in whatever they are obtained and the final reasoned out account of the result shown by the statistics." **— Bowley**

"A statistical table is a systematic organisation of data in column and rows." **— Prof. Neiswanger**

"A statistical table is a classification of related numerical facts in vertical columns and horizontal rows." **— W.A. Spurr and C.P. Bonini**

"A statistical table is a presentation of numbers in a logical arrangements with some brief explanation to show what they are." **— C.T. Clarks and L.L. Schkade**

"Tabulation is the process of condensing classified data in the form of a table so that it may be more easily understood and so that any comparisons involved may be more readily made." **— D. Greogory and H.Ward**

3.1.2 Objectives of Tabulation

The main objective of statistical table is to summarise the huge data in a simple and systematic way so as to be easily understood, analysed and interpreted for specific purposes.

The main objectives of tabulation are :

(1) To clarify the purpose of Investigation.

(2) To bring out essential features of the data.

(3) To present numerical data in an attractive form.

(4) To present data in a minimum space.

(5) To facilitate comparison of data.
(6) To detect errors and omissions in the data.
(7) To facilitate reference to the data with a suitable title and a number so that it becomes distinct and comparable.
(8) To identify trend and tendencies of the given data.
(9) To facilitate statistical processing.

3.1.3 Difference between Classification and Tabulation

Classification and tabulation are closely related to each other, but still there are some differences between their processes. They are :

(1) Both the process have follow a sequence. First, data are classified and then they are presented in tables. Neither of them is substitute for the other. Hence classification lay a basis for tabulation.
(2) In classification, data are classified into different classes according to their similarities and dissimilarities whereas in tabulation, classified facts are places in rows and columns under suitable headings. Thus tabulation is a mechanical function of classification.
(3) Classification is preliminary to tabulation and prepares the ground for proper presentation of statistical data. It is only after proper tabulation that statistical analysis and interpretation are possible.
(4) Classification is a process of statistical analysis, tabulation is a process of presenting data in suitable manner.

3.1.4 Advantages of Tabulation

The tabular presentation of data has several advantages :

(1) It helps in presenting the complex data clearly and easily be understood. After tabulation, data get a definite form and structure, which can be studied from left to right and top to bottom easily. It reveals patterns within the data.
(2) It highlights the significant and relevant details of the information in the data.
(3) It enables to draw the attention of the user to specific observations by means of comparisons, emphasis and arrangement of the layout.
(4) It also facilitates the detection of the errors and the emissions in the data.
(5) It occupies less space and hence save paper and time.
(6) It facilitates computation of statistical data as various methods of calculation like average, dispersion, correlation, regression, time series etc. require data must be presented in tables.
(7) Tabulation is helpful in classification too. Classification is a theoretical process of data analysis and tabulation is an applied process of it. So while classifying data, the characteristics divided in different classes are not clear till they are placed in a suitable table.
(8) Tabulated data makes the data suitable for further diagrammatic and graphic representation.

3.1.5 Limitations of Tabulation

Tabulation has certain limitations too.

These are :

(1) Tables contain only numerical data but not their details and descriptions. So it is not possible to understand it easily.

(2) It is not possible to present qualitative facts in tables.

(3) Individual characteristics of data end in tabulation.

(4) It requires a specialised knowledge to understand a table, so a layman cannot derive any conclusion from a table.

3.1.6. Components of a Table

The components (or parts) of a table depends upon nature and purpose of data. However for a good statistical table, the following components are must :

(1) **Table Number :** Numbering of tables in a logical sequence is necessary for proper identification and ready reference at the time of need. It is must when there are a large number of tables under study. Table number should be given at the top of the table either to the centre above the title or in the side of the title.

(2) **Title :** Every table must have a title and it should be placed at the top of it. Title should be simple, brief, concise and self explanatory. It should briefly describe the subject matter under study, the time period, the place of study, the basis of classification of data, the sources of information etc.

(3) **Captions and Stubs :** Headings or sub-headings used to describe columns or sub-columns in the upper part of the table are called ***captions,*** while in the extreme left part of the table, where descriptions of rows are shown are called **stubs.** A caption or a stub should be brief and self-explanatory.

(4) **Head Notes :** Head note is a sort of supplement to the title to provide an explanation for all or a major part of the material in the table. It is placed just below the title and enclosed in brackets. For example, units of measurements is usually written as a head note such as *'in thousands', 'in millions', 'in rupees',* etc.

(5) **Body :** The body of the table is the central part of table that contains the numerical information presented in the table. It is the most important part of the table. The arrangement in the body is generally from left to right in the rows and from top to bottom in the columns. Each item in the body is called ***'cell'.***

(6) **Foot Note :** Foot notes are used only if feature or item of the table has not been adequately explained, or need some special attention, or some additional information is required etc. A footnote, if any, placed at the bottom of the table directly below the body of the table. Footnotes are also necessary to specify important limitation of data, if any.

(7) **Source Note :** If the source of the data is secondary in nature, then source note must be given at the bottom of the table.

It is important for few reasons as :

(a) It gives credit to the individual or group who collected the data.

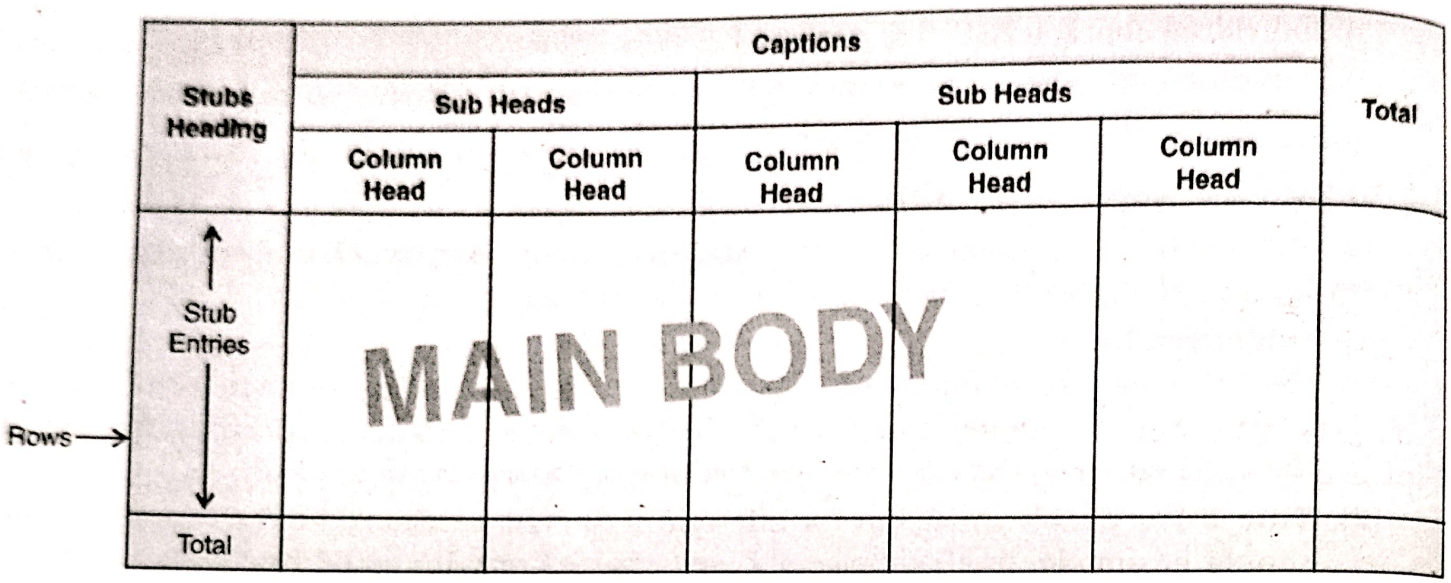
(b) It provides a hint about the source material, which may be more useful and complete.

(c) It increases the reliability and accuracy of the data given in the table by referring to the original source.

STRUCTURE OF A TABLE

Title (Head Note) **Table Number**

Columns



Foot Note :
Source :

(I) On the Basis of Construction statistical tables can be divided as :

(a) Simple Table

(b) Complex Table

(a) Simple Table : If the data is presented in table on the base of one characteristic only then the table is known as simple table or ***one-way table or a table of the first order***. Such table is easy to construct and understand. The following is an example of a simple table.

Example 1.

Table : No. of students in different classes in P.U., Year 2000

Class	*No. of Students*
M.A.	700
M.Sc.	500
M.Com.	200
B.A.	1000
B.Sc.	700
B.Com.	400
Total	3500

(b) Complex Table : In complex table, two or more characteristics of the data are presented. These tables are frequently used in practice because they enable full information to be incorporated and facilitate a proper consideration of all related facts. It may be

(i) DOUBLE TABLE : If the data are tabulated on the basis of only two characteristics, then the table is known as double table or ***two way table.***

Example 2.

Table : P.U

No. of students according to class and sex, in year 2000

Class	No. of Students		Total
	Boys	Girls	
M.A.	400	300	700
M.Sc.	200	300	500
M.Com.	100	100	200
B.A.	400	600	1000
B.Sc.	350	350	700
B.Com.	250	150	400
Total	1700	1800	3500

(ii) TREBLE TABLE : In such a table, the data are tabulated on the basis of three interrelated characteristics and is also known as ***three-way table.***

Example 3.

Table : P.U

No. of students according to class, sex and employment, year 2000.

Class	No. of Students				Total		
	Boys		Girls				
	Employed	Unemployed	Employed	Unemployed	Employed	Not Employed	Total
M.A.							
M.Sc.							
M.Com.							
B.A.							
B.Sc.							
B.Com.							
Total							

(iii) MANIFOLD TABLE : In such a table, the data are tabulated on the basis of a large number of interrelated characteristics and is also known as ***Higher order table or Many-way table.***

In complex tables, the columns of tables are further divided into sub-columns.

Example 4.

Table : P.U

No. of students according to class, age, sex and Residence.

Class	Age	No. of Students				Total		
		Boys		Girls				
		Hostler	Day-Scholar	Hostler	Day Scholar	Hostler	Day-Scholar	Total
1. Arts	15 – 20							
	20 – 25							
	Above 25							